

Arbitrage and the Logic of Pricing

Financial Engineering · Section 1

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Outline

- 1.1 The pricing paradigm
- 1.2 The one-period market model
- 1.3 State prices and the fundamental theorem
- 1.4 The stochastic discount factor and risk-neutral probabilities



The Pricing Paradigm

Two kinds of pricing

- **Equilibrium pricing** asks what an asset is worth in absolute terms. It models preferences, endowments and beliefs, clears supply against demand, and reads prices off the equilibrium.
- **Relative pricing** asks a narrower and far more robust question. Taking some prices as *given* by the market, what must a related asset cost so that the two are consistent?
- Financial engineering is relative pricing. We rarely ask why the stock is worth \$100. We ask what an option on it must be worth, given that it is.

Replication, and what enforces it

- A portfolio of traded assets that reproduces a target payoff state by state is the same future cash flow as the target.
- Two claims to the identical cash flow must trade at the identical price today. That is the **law of one price**.
- So the target's price is forced: it is the cost of building the replicating portfolio.
- What guarantees the law of one price is the absence of **arbitrage**. Every pricing formula in this course is, at bottom, the cost of a replicating strategy.

Example 1.1

Example 1.1

Using only the bond $(1, 1)$ priced at 0.80 and the stock $(2, 1)$ priced at 1.40, find the price of a structured claim X paying \$4 in state 1 and \$3 in state 2.

- No preferences, no probabilities, no discounting assumption enters.



The One-Period Market Model

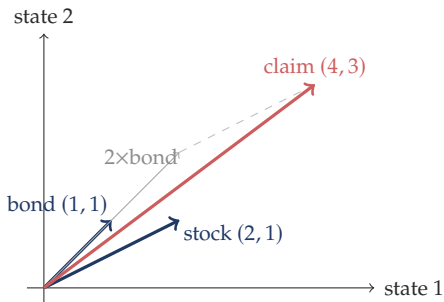
The market

Definition 1.2 (One-period market model)

N traded assets. Asset j has price $p_j \in \mathbb{R}$ today and pays D_{sj} in state s ; $D = (D_{sj})$ is the $S \times N$ *payoff matrix*. A portfolio $\theta \in \mathbb{R}^N$ costs $p^\top \theta$ and pays $D\theta$. A payoff x is *marketed* if $x = D\theta$ for some θ , and $\mathcal{M} = \text{range}(D)$ is the *marketed subspace*.

- $\mathbb{R}^S$ holds every conceivable payoff. Trading manufactures exactly those in $\mathcal{M}$.
- Prices are not assumed positive: a forward costs nothing to enter.
- $\mathcal{M}$ comes from D alone, never from p . What the market can **build** and what it **charges** are separate questions.

The state-payoff plane



- A portfolio is a linear combination of the traded assets. Here the claim is $2 \times \text{bond}$ plus stock, which is Example 1.1.
- Two independent assets span the plane, so every payoff is marketed.

Arbitrage

Definition 1.3 (Arbitrage)

A portfolio θ of either type:

- **Type I:** $p^\top \theta = 0, D\theta \geq 0, D\theta \neq 0$.
- **Type II:** $p^\top \theta < 0$ and $D\theta \geq 0$.

A market is *arbitrage-free* if it admits neither.

- One idea, two sides. Type I takes value *tomorrow* for free, type II takes it *today*.
- Given an asset with positive price and non-negative, non-zero payoff, type II converts to type I: park the cash there. Without one it need not, so we exclude both.

Redundancy

Definition 1.4 (Redundant asset)

An asset is *redundant* if its payoff is a linear combination of the other assets' payoffs. A market has *no redundancy* when the columns of D are linearly independent.

- A redundant asset adds nothing to $\mathcal{M}$. It may be exactly the instrument we want to price, and no-arbitrage leaves its price no freedom.

The law of one price

Proposition 1.5

In an arbitrage-free market, two portfolios with the same payoff have the same cost: $D\theta = D\theta'$ implies $p^\top \theta = p^\top \theta'$. In particular a redundant asset's price is the cost of any portfolio that replicates it.

- With no free lunch, the price of a payoff depends only on the payoff. Anything you can build must cost what the parts cost.

Example 1.6

Example 1.6

Three assets over two states: a bond $(1, 1)$ at 0.80, a stock $(2, 1)$ at 1.40, and a structured note $(4, 3)$ at 2.80. Is the market arbitrage-free? If not, construct an explicit arbitrage.

State Prices and the Fundamental Theorem

State prices

Definition 1.7 (State-price vector)

$\psi = (\psi_1, \dots, \psi_S)^\top$ is a *state-price vector* if it is strictly positive, $\psi \gg 0$, and reproduces every traded price:

$$p_j = \sum_{s=1}^S \psi_s D_{sj} \quad \text{for every } j, \quad \text{that is} \quad p = D^\top \psi.$$

- ψ_s is what the rule charges for \$1 delivered if and only if state s occurs.

Is ψ_s a market price?

- Not necessarily. Read ψ_s as an **implicit** state price: a value assigned by a positive linear rule that reproduces every observed price.
- If the Arrow security for state s is **attainable**, then ψ_s is its unique no-arbitrage price.
- If it is not attainable, Theorem 1.11 says several admissible ψ exist, and they can value it differently.
- Same caution for the riskless rate: $\sum_s \psi_s = 1/(1+r)$ needs the payoff $(1, \dots, 1)$ to be traded or replicable. Otherwise there is no single r to read off.

What the law of one price already gives

Proposition 1.8 (Law of one price and linear pricing)

The law of one price holds **if and only if** some vector $\psi \in \mathbb{R}^S$, with signs unrestricted, satisfies $p = D^\top \psi$.

- So the law of one price already buys a linear pricing rule. It does not make that rule positive.
- Bond (1, 1) at 1.00 and stock (2, 1) at 0.50: the payoffs are independent, so the law of one price holds automatically. But $\psi = (-0.5, 1.5)$, and shorting one bond against one stock costs -0.50 and pays $(1, 0)$. A free lunch.
- Positivity is exactly what no-arbitrage adds. That is the next theorem.

The one imported result

Separating hyperplane theorem (imported, not numbered)

Let $M \subseteq \mathbb{R}^n$ be a linear subspace and $K \subseteq \mathbb{R}^n$ non-empty, compact and convex, with $M \cap K = \emptyset$. Then there is $\lambda \in \mathbb{R}^n$ with $\lambda^\top z = 0$ for all $z \in M$ and $\lambda^\top k > 0$ for all $k \in K$.

- A subspace that never meets a compact convex set can be separated from it by a plane through the origin, with the whole set strictly on the positive side.
- This is the only result in the section imported without proof, and it is used here and nowhere else in the course. The normal λ becomes the prices.

The fundamental theorem

Theorem 1.9 (FTAP, one period)

A one-period market admits no arbitrage **if and only if** there is a strictly positive state-price vector ψ with $p = D^\top \psi$.

- A market has no free lunch exactly when there is a consistent set of positive prices for state-by-state \$1 payoffs that reproduces every quoted price.
- This is the theorem the whole course rests on. Everything later is this statement on a richer probability space.

Complete markets

Definition 1.10 (Complete market)

The market is *complete* if every payoff is marketed, $\mathcal{M} = \mathbb{R}^S$; equivalently $\text{rank}(D) = S$. Otherwise it is *incomplete*.

- Completeness says every contingent claim can be replicated: every risk perfectly hedged, every derivative exactly manufactured.
- It needs at least as many independent assets as states, so $N \geq S$.

What completeness buys

Theorem 1.11 (Completeness and uniqueness)

In an arbitrage-free market, the market is complete **if and only if** the state-price vector is unique.

- If the market can build every payoff, the state prices are pinned down to a single vector. If some payoff cannot be built, many state-price vectors price the traded assets equally well.
- So no-arbitrage alone does not price a claim outside $\mathcal{M}$.

Three conditions, one object

Market condition	Equivalent statement about ψ
Law of one price	some $\psi \in \mathbb{R}^S$ solves $p = D^\top \psi$
No arbitrage	some $\psi \gg 0$ solves $p = D^\top \psi$
No arbitrage and completeness	exactly one ψ solves $p = D^\top \psi$

- Each row adds one word. A pricing rule, then a *positive* one, then a *unique* one.
- The three are not equivalent to each other. No-arbitrage implies the law of one price and not the reverse. Completeness neither implies nor is implied by either of the other two.

Example 1.12

Example 1.12

- (a) Bond $(1, 1)$ at 0.80 and stock $(160, 60)$ at 100, two states. Find the state-price vector.
- (b) Now three states, same two assets: stock $(160, 100, 60)$ still at 100, bond $(1, 1, 1)$ at 0.80. Find *all* state-price vectors and describe the set geometrically.

The Stochastic Discount Factor and Risk-Neutral Probabilities

Why one discount rate cannot work

Market of Example 1.12(a): $\psi = (0.52, 0.28)$, so $r = 25\%$. Let the physical probabilities be $\pi = (0.8, 0.2)$.

- Stock's expected payoff is $0.8(160) + 0.2(60) = 140$. Discounted at the riskless rate that is $140/1.25 = 112$. The stock trades at 100.
- Discount the stock at its own rate instead? That rate is $140/100 = 1.40$. But the bond needs $1/0.80 = 1.25$. Every asset would need a rate of its own, so that is no rule at all.
- So the correction cannot be one number. It must depend on the **state**.

Solving for the state-by-state correction

Find m_1, m_2 that reweight the payoff inside the expectation, so one formula prices both traded assets: $\pi(X) = \pi_1 m_1 X_1 + \pi_2 m_2 X_2$.

- Bond and stock give $0.8m_1 + 0.2m_2 = 0.80$ and $0.8m_1(160) + 0.2m_2(60) = 100$.
- With $u = 0.8m_1, v = 0.2m_2$: $u + v = 0.80$ and $160u + 60v = 100$, the state-price system of Example 1.12(a) again.
- So $u = 0.52, v = 0.28$, and $m_1 = \frac{0.52}{0.8} = 0.65, m_2 = \frac{0.28}{0.2} = 1.40$.
- The correction in each state is the state price divided by the physical probability.

The stochastic discount factor

Definition 1.13 (Stochastic discount factor)

Given physical probabilities $\pi \gg 0$,

$$m_s = \frac{\psi_s}{\pi_s}, \quad s = 1, \dots, S,$$

is the *stochastic discount factor*, or pricing kernel.

- Since $\psi_s = m_s \pi_s$, every price is a physical expectation: $\pi(X) = \mathbb{E}[mX]$. Setting $X = (1, \dots, 1)$ gives $\mathbb{E}[m] = 1/(1+r)$.
- Its variation across states is what prices risk.

What no-arbitrage does not tell you about m

- Here $m_2 = 1.40$ against $m_1 = 0.65$, so the kernel is larger in the low-payoff state.
- But no-arbitrage delivers only a **positive** m . It does not say which state is bad, and it does not force m to be large there. That $m_2 > m_1$ is just what these prices imply.
- The familiar reading needs economic structure. In a consumption-based model m is the marginal rate of substitution, so m is high in bad states, where marginal utility is high.
- Then a claim paying off in bad states is expensive. The intuition comes from that model, not from Theorem 1.9.

The risk-neutral measure

Definition 1.14 (Risk-neutral measure)

Divide each state price by the total $\sum_u \psi_u$ instead of by π_s :

$$q_s = \frac{\psi_s}{\sum_u \psi_u} = (1 + r) \psi_s, \quad s = 1, \dots, S.$$

By construction $q_s > 0$ and $\sum_s q_s = 1$, so Q is a genuine probability distribution on the states.

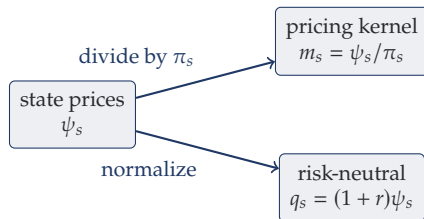
- Substituting $\psi_s = q_s/(1 + r)$ gives $\pi(X) = \mathbb{E}^Q[X]/(1 + r)$.
- The kernel keeps the real probabilities and corrects the payoff; Q leaves the payoff alone and corrects the probabilities.

How far does $\mathbb{E}^Q[X]/R_f$ reach?

Market	Risk-neutral measure	What a claim gets
Complete	exactly one Q	one price, every claim
Incomplete	many admissible Q	one price if marketed, else an interval

- If $X = D\theta$ is **marketed**, then $\pi(X) = \psi^\top D\theta = p^\top \theta$ for *every* admissible ψ . All the Q 's agree, and the answer is the cost of the replicating portfolio.
- If X is not marketed, different admissible Q give different numbers, and no-arbitrage does not choose between them.
- This is the payoff of §1.3: completeness is what turns the pricing rule from a range into a number.

One object, three faces



- $\pi(X) = \sum_s \psi_s X_s = \mathbb{E}[mX] = \frac{\mathbb{E}^Q[X]}{1 + r}$.
- $q_s / \pi_s = m_s / \mathbb{E}[m]$: the kernel, rescaled to average one, is the ratio of risk-neutral to physical probabilities.

Example 1.15

Example 1.15

Complete market of Example 1.12(a): bond $(1, 1)$ at 0.80, stock $(160, 60)$ at 100, state prices $\psi = (0.52, 0.28)$, physical probabilities $\pi = (0.8, 0.2)$.

(a) Find r and verify $\mathbb{E}[m] = 1/(1+r)$ for $m = (0.65, 1.40)$. (b) Price the call struck at $K = 100$, payoff $(60, 0)$, using the kernel, and check the answer by explicit replication. (c) Find Q and price the call again under Q . (d) Show the call price is unchanged if π becomes $(0.5, 0.5)$.